

Eduardo de Jesús Dávila Meza, Ph.D.

AI/ML & DL • Computer Vision • Embedded/Robotics Systems (ROS/ROS2)

✉ eduardodavila94@hotmail.com | ☎ +52 33 2969 2743 | 📁 Professional Portfolio | 🏠 [EduardoDavilaMeza](#) | 🐼 [eDavila-DrRaccoon](#)
🎓 Federal Professional Certificates: Bachelor's Degree: 12027207 | Master's Degree: 14043743 | Ph.D. Degree: 15067339

Engineering Products

Vision Pipeline – Production-oriented Computer Vision Inference Framework | Modular FastAPI/Docker inference system using YOLO11 and PyTorch, with REST/OpenAPI interfaces, YAML configuration, structured logging, CLI inference, JSON response schema, automated testing, CI, and reproducible benchmarking.

🔧 Python • PyTorch • YOLO11 • FastAPI • Docker • OpenAPI • pytest • GitHub Actions

ROS/ROS2 Robotics – Robotics Software & Perception Systems | Modular ROS/ROS2 repositories covering differential-drive-robot programming, simulation, computer vision, navigation, embedded Linux and deployment-oriented robotics workflows.

🔧 Stack: C++ • Python • ROS/ROS2 • OpenCV • Linux • Gazebo • RViz • Bash

Technical Skills

AI/Computer Vision: Computer Vision, Deep Learning (Mask R-CNN), Object Detection, Image Segmentation, Model Optimization, PyTorch, TensorFlow/Keras, OpenCV

Software Engineering: Python, C++, SQL, Bash, REST APIs, FastAPI, OpenAPI, pytest, JSON, XML, YAML

Deployment/Infrastructure: Docker, Docker Compose, Git, GitHub, GitHub Actions

Robotics/Embedded: ROS/ROS2, RViz, Gazebo, Embedded Linux, C++/Python robotics development

Data/ML: NumPy, Pandas, Scikit-learn, Matplotlib, SQLAlchemy, Time Series Analysis (ARIMA), LSTM

GNU/Linux: Ubuntu & Arch Linux

Documentation: LaTeX, Markdown, MediaWiki, MS Office, Dia Diagram

Engineering Focus

- Computer vision and deep-learning systems
- Modular AI inference and deployment architectures
- ROS/ROS2 perception and robotics systems
- Production-oriented Python/C++ development
- API, testing and reproducible deployment workflows

Spoken Languages

Spanish | Native: full professional proficiency.
English | B2 Advanced / Professional Working Proficiency.
Certified by Cinvestav and Desafío Latam.

Work Experience & Projects | Full List

Robot Programming Instructor – ROS/ROS2 April 2024 – Present
Tecnológico de Monterrey (ITESM), Guadalajara
Zapopan, Jalisco

1) Designing and delivering hands-on courses in ROS/ROS2 using C++ and Python, focused on differential drive robots. 2) Developing practical repositories and lecture notes covering the foundations of development, simulation, deployment, and monitoring of ROS/ROS2-based modular systems, including computer vision and autonomous navigation. 3) Recognized as a top-rated instructor (See recognition). See repository 2025. See latest repository.

🔧 C++ • Python • ROS/ROS2 • Bash • Computer Vision • Robotics • Embedded Linux • Technical instruction

Mathematical Analyst in Information Technologies September 2025 – April 2026
Geovoy, Busmen Group
Tlajomulco de Zúñiga, Jalisco

1) Built an automated reporting tool using LLM-powered APIs to generate analytical reports and insights. 2) Developed statistical and machine learning models (time series analysis (ARIMA), LSTM) for route optimization and demand prediction, generating insights to improve operational decision-making. 3) Designed and optimized APIs using FastAPI, raw SQL, and Object-Relational Mapping (ORM) to integrate statistical models into production systems for geolocation and inventory management. 4) Deployed a MediaWiki-based platform to standardize internal processes, code, and API documentation.

🔧 FastAPI • SQLAlchemy • Time Series • LSTM • Forecasting • Predictive Modeling • GPT-OSS-20B • Ollama • Data Analysis • Bash

AI/Deep Learning Research Engineer – Computer Vision May 2021 – July 2024
Cinvestav, Guadalajara: Recognition of Fundus Pathologies – Medical Image Segmentation
Zapopan, Jalisco

1) Developed a deep learning model for medical image analysis, in collaboration with German research institutions, to support clinical decision-making, focusing on segmentation and quantification of retinal pathologies. 2) Applied data preprocessing, augmentation, feature extraction, and model validation techniques to ensure robustness and reliability in healthcare-related datasets. 3) Worked with high-resolution image datasets to improve the detection and segmentation of retinal pathologies. 4) Contributed to a research publication in computer vision and medical imaging.

🔧 Computer Vision • Deep Learning • TensorFlow/Keras • Mask R-CNN • Image Segmentation • Data labeling, augmentation, and visualization

Computer Vision & ROS Developer December 2019 – October 2023
Cinvestav, Guadalajara: Intelligent Visual Guide System (OJO SMART) – Modular Navigation Device
Zapopan, Jalisco

1) Developed ROS-based modules for real-time recognition of colors, objects, signs, banknotes, and text using computer vision techniques. 2) Contributed to the integration of multiple perception systems into an assistive navigation device for visually impaired users.

🔧 Computer Vision • C++ • Python • ROS • Tesseract OCR • OpenCV • TensorFlow • Embedded Linux

Publications & Patents

Meeting Abstract | June 2024 | “Deep-learning based quantification of RPE65-mutation inherited retinal degeneration from wide-angle images – novel biomarkers for natural history and interventional studies”, at *Investigative Ophthalmology & Visual Science*, vol. 65(7), 1392, ID: 2794864.

🔧 AI/ML lifecycle • Computer vision / Medical image analysis • Data visualization • Mask R-CNN • Feature extraction • Research

Journal Article | September 2023 | “Quaternion and Split Quaternion Neural Networks for Low-Light Color Image Enhancement”, at *IEEE Access*, vol. 11, pp. 108257-108280, 10.1109/ACCESS.2023.3312234.

🔧 AI/ML lifecycle • Computer vision / Image color analysis • Quaternion algebras • Color spaces • EKF

Patent | March 2017 | “Device for controlling underactuated two-link systems with one actuator”, filed under the Invention Support Program, University of Guadalajara. Application no. MX/a/2017/016436.

🔧 Embedded systems • Control theory • Digital and power electronics • PICs • SPI & I2C communication protocols

Academic Degrees

Ph.D. in Science – AI/Deep Learning | Cinvestav, Guadalajara September 2019 – May 2024

Thesis | Deep learning for recognition and quantification of fundus pathologies using instance segmentation, and quaternion neural networks for low-light image enhancement. Download thesis.

🔧 AI/ML lifecycle • Computer vision / Image analysis • Deep Learning • CNN/NN • Robotics • Research • Science Communication

M.Sc. – AI/Machine Learning | Cinvestav, Guadalajara September 2017 – August 2019

Thesis | Quaternion neural networks for low-light image enhancement, and identification of an electromechanical system. Download thesis.

🔧 AI/ANNs • Computer vision • Linear algebra • Control theory • Robotics • Probability & statistics • Research • Science Communication

B.Eng. in Mechatronics – Embedded Systems | University of Guadalajara August 2012 – December 2016

Social Service & Professional Internship | Developed embedded and mechatronic projects in an electronics and telecommunications laboratory.

🔧 Embedded systems • Linear algebra • Calculus • Control theory • Digital and power electronics • HMI • PICs & Arduino

Certifications | Additional Training

AI/DL
Python

AI/ML
R for DS

Data Cleaning
ROS

LLMs
Scikit-learn

NLP
SQL

OpenCV
TensorFlow/Keras

Pandas
Full list